

Hospital & Clinical Management System

Hospital Management System

(SQL Server)

A complete database solution for managing clinical operations, patient admissions, billing, and lab diagnostic tests, designed in accordance with the project specification report.

Project Overview

This relational database models core healthcare workflows:

- **Core Clinical Operations:** Patients, Doctors, Appointments, Prescriptions, and Medications.
- **Inpatient Management:** Room assignments, stay tracking, and admission billing.
- **Custom Laboratory Module:** Diagnostic test catalog (`LabTest`) and result entries (`LabResult`) with range validations.
- **Automated Business Rules:** Conflict-free booking, auto-updating room availability, payment auditing, and clinical history preservation.

Programmability & Database Architecture

1. Database Schema (DDL)

Includes tables with integrity constraints (`PRIMARY KEY`, `FOREIGN KEY`, `CHECK`, `UNIQUE`):

- Doctors, Patients, Appointments, Prescriptions, Medications, PrescriptionDetails, Rooms, Admissions, Bills, LabTests, LabResults, PaymentAuditLog.

2. Database Views (`vw_`)

- **`vw_TodayAppointments`:** Overview of appointments scheduled for the current date.
- **`vw_PatientBills`:** Summary of patient invoices and payment statuses.
- **`vw_ActiveRoomAdmissions`:** Currently admitted patients and occupied rooms.
- **`vw_DoctorWorkload`:** Doctor metrics and assigned appointment counts.
- **`vw_RevenueBySpecialty`:** Total paid revenue grouped by medical specialty.

3. User-Defined Functions (fn_)

- **fn_GetPatientAge:** Calculates patient age based on birth date.
- **fn_GetPatientTotalPaid:** Computes total paid amount for a given patient.
- **fn_CalculateAdmissionCost:** Calculates stay duration and total cost.
- **fn_GetPatientMedicalHistory:** Inline TVF returning patient history.
- **fn_GetDoctorSchedule:** Inline TVF listing non-cancelled doctor appointments.

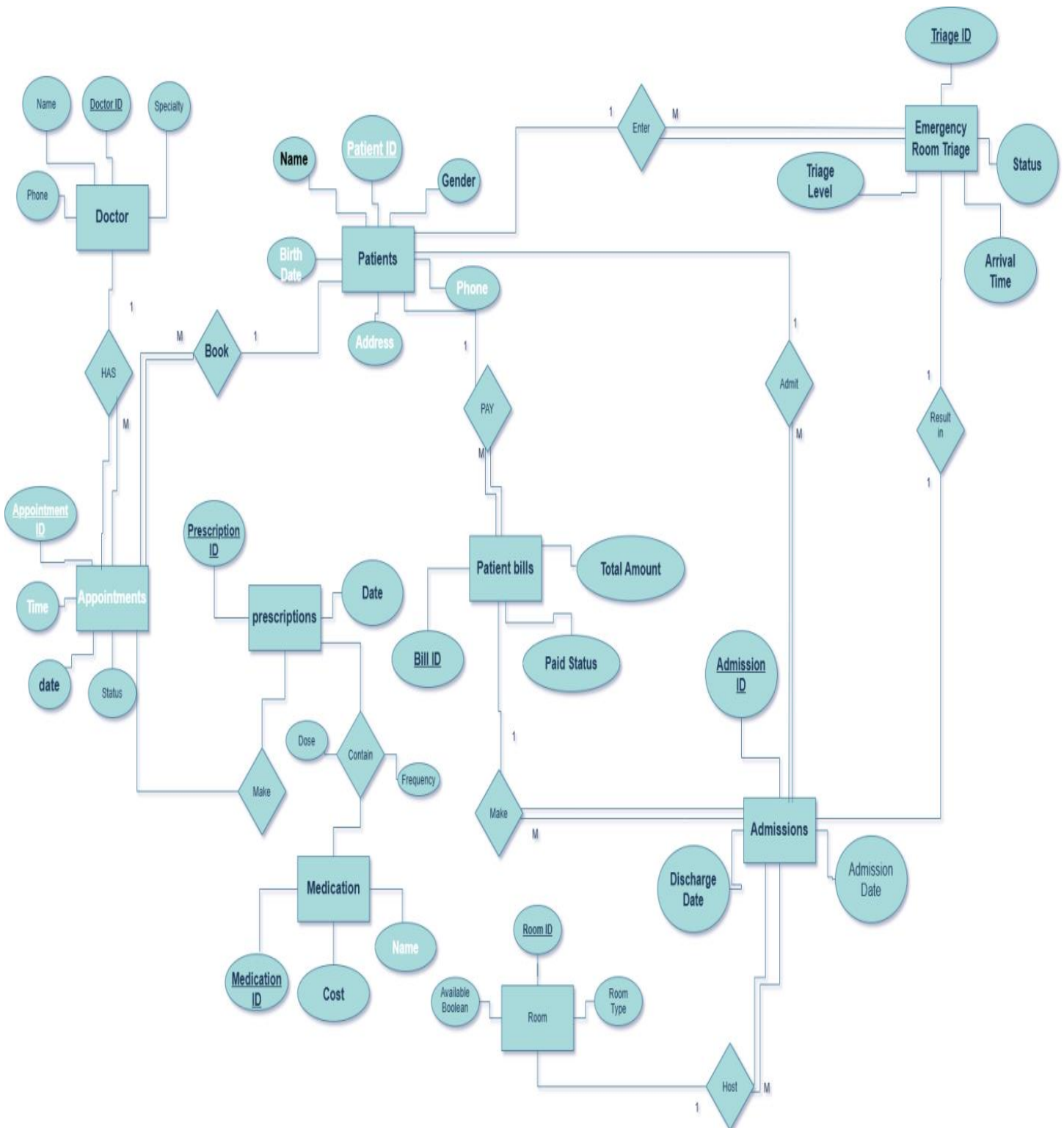
4. Stored Procedures (sp_)

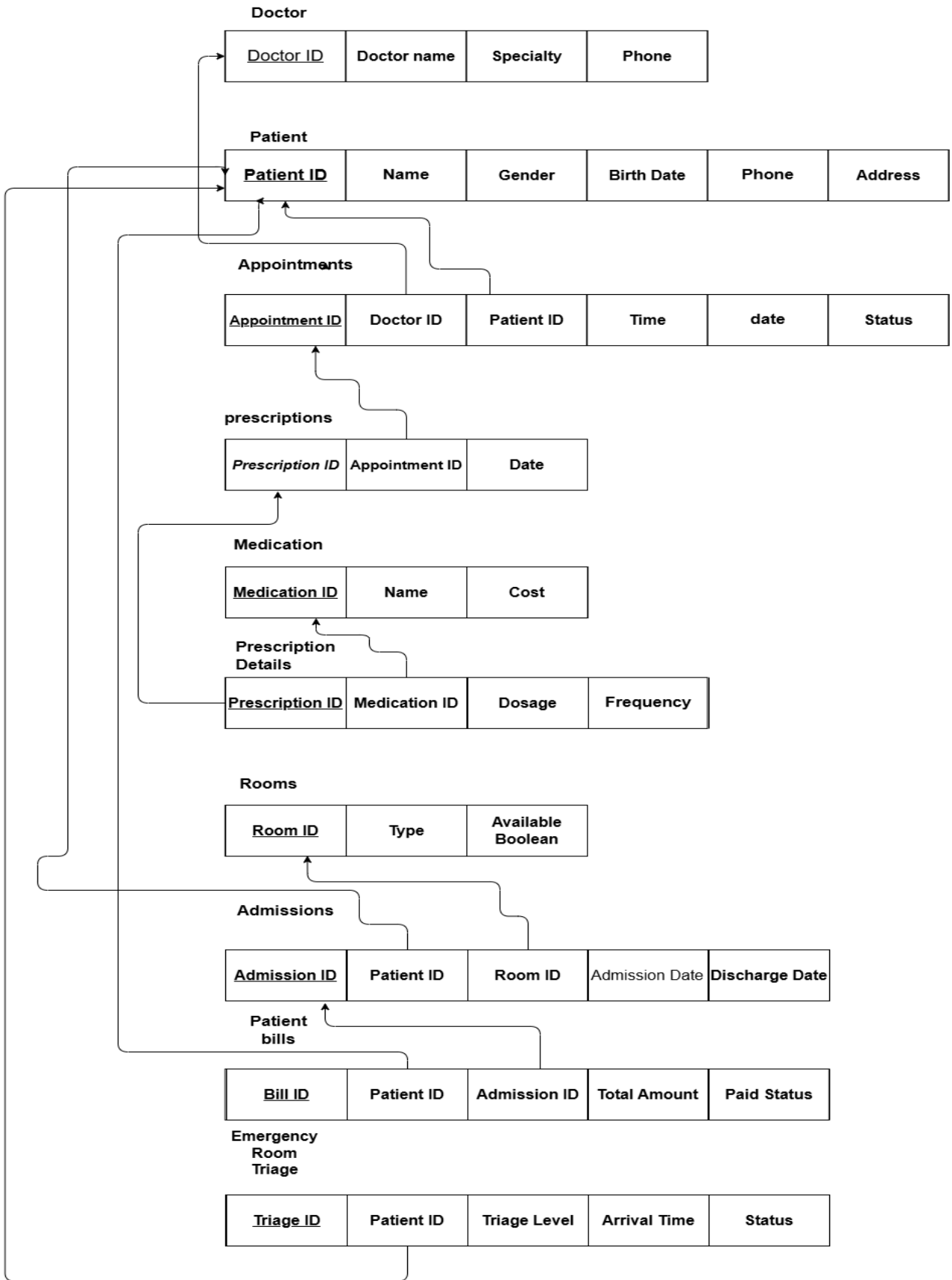
- **sp_BookAppointment:** Adds new appointments.
- **sp_AdmitPatient:** Registers inpatient admissions.
- **sp_GenerateBill:** Computes room, medication, and lab costs to generate a bill.
- **sp_DischargePatient:** Discharges an inpatient and generates their bill in a transaction.
- **sp_ProcessBillPayment:** Processes bill payments with validation.

5. Database Triggers (trg_)

- **trg_PreventDoubleBooking:** Prevents overlapping doctor appointment slots.
- **trg_CheckRoomAvailability:** Blocks admissions into occupied rooms.
- **trg_AutoFreeRoomOnDischarge:** Frees room availability when `DischargeDate` is updated.
- **trg_PreventMedicationDelete:** Blocks deleting medications with active prescriptions.
- **trg_AuditBillPayment:** Logs payment details to `PaymentAuditLog` upon payment completion.

System Architecture & Database Documentation





- **Core Entities:** Manages central hospital entities including `Doctor`, `Patient`, `Appointments`, `prescriptions`, `Medication`, `Prescription Details`, `Rooms`, `Admissions`, and `Patient bills`.
- **Custom Module (Emergency Room Triage):** Extended with `Emergency Room Triage` table to manage emergency patient intake, priority scoring (`Triage Level`), arrival tracking (`Arrival Time`), and triage status (`Status`).
- **Integrity Constraints:** Primary keys and foreign keys maintain strict relational integrity across all clinical, prescription, admission, billing, and emergency modules.

Seed Data Population & Record Metrics

	TableName	TotalRecords
1	Patients	5
2	Doctor	4
3	Appointments	5
4	Room	5
5	Admissions	3
6	Medication	5
7	Prescriptions	3
8	PatientBills	4
9	EmergencyRoomTriage	4

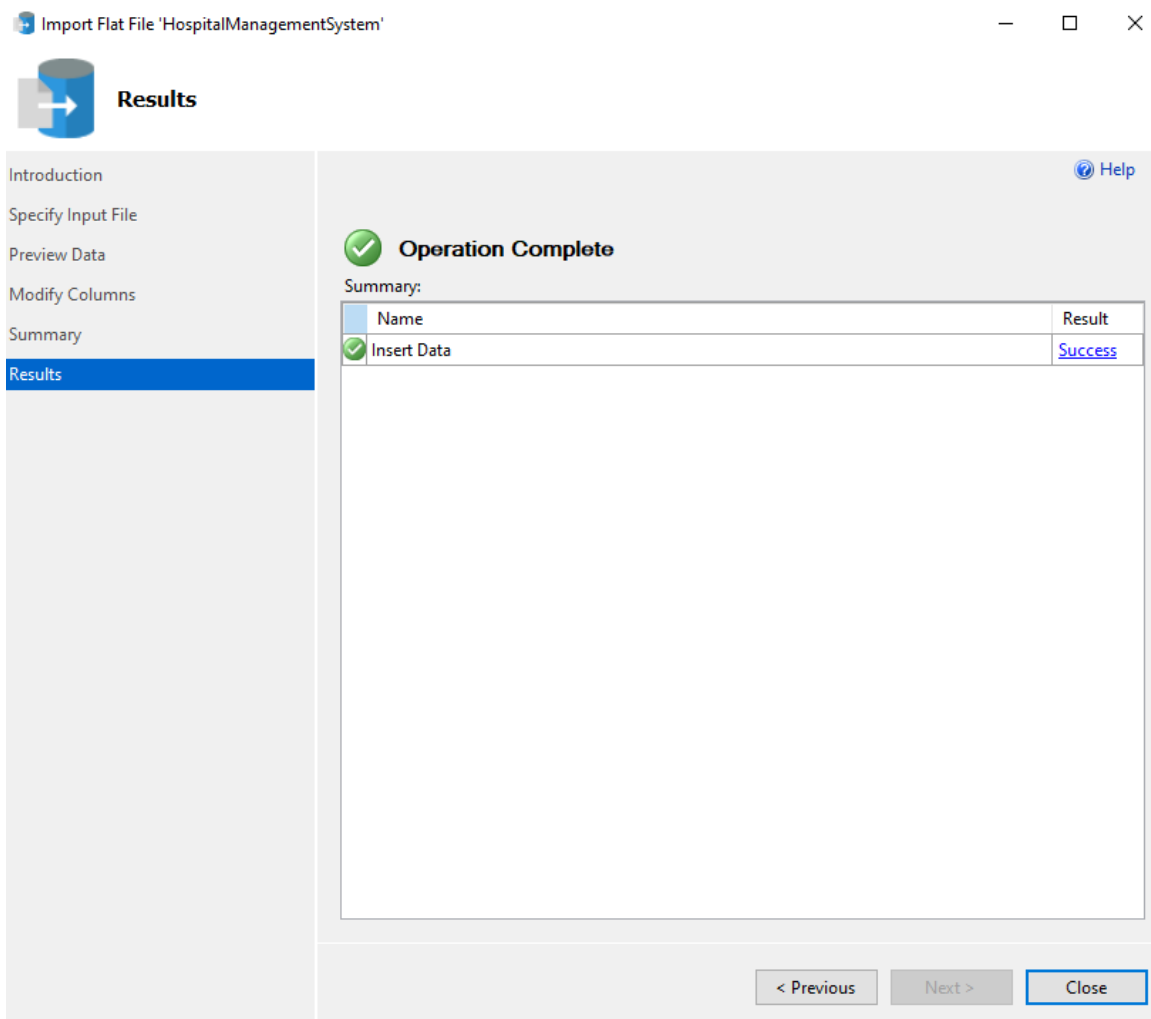
Programmability Objects Inventory

	Object Type	Object Name
1	SQL_SCALAR_FUNCTION	fn_diagramobjects
2	SQL_STORED_PROCEDURE	sp_AddPatientToTriage
3	SQL_STORED_PROCEDURE	sp_AdmitPatient
4	SQL_STORED_PROCEDURE	sp_alterdiagram
5	SQL_STORED_PROCEDURE	sp_BookAppointment
6	SQL_STORED_PROCEDURE	sp_creatediagram
7	SQL_STORED_PROCEDURE	sp_CreatePrescriptionWithMedication
8	SQL_STORED_PROCEDURE	sp_DischargePatient
9	SQL_STORED_PROCEDURE	sp_dropdiagram
10	SQL_STORED_PROCEDURE	sp_GenerateBill
11	SQL_STORED_PROCEDURE	sp_helpdiagramdefinition
12	SQL_STORED_PROCEDURE	sp_helpdiagrams
13	SQL_STORED_PROCEDURE	sp_ProcessBillPayment
14	SQL_STORED_PROCEDURE	sp_renamediagram
15	SQL_STORED_PROCEDURE	sp_upgraddiagrams
16	SQL_TRIGGER	trg_PreventDoubleBooking
17	VIEW	vw_ActiveRoomAdmissions
18	VIEW	vw_DoctorWorkload
19	VIEW	vw_PatientBillingSummary
20	VIEW	vw_PatientBills
21	VIEW	vw_RevenueBySpecialty

- **Views:** Reporting views for workload, active admissions, today's appointments, and specialty revenue (vw_TodayAppointments, vw_PatientBills, vw_ActiveRoomAdmissions, vw_DoctorWorkload, vw_RevenueBySpecialty).
- **User-Defined Functions:** Scalar and Table-Valued functions for medical histories, age, total payments, and schedules (fn_GetPatientAge, fn_GetPatientTotalPaid, fn_CalculateAdmissionCost, fn_GetPatientMedicalHistory, fn_GetDoctorSchedule)
- **Stored Procedures:** Business workflow procedures (sp_BookAppointment, sp_AdmitPatient, sp_DischargePatient, sp_GenerateBill, sp_ProcessBillPayment).

- **Triggers:** Automated integrity rules (trg_PreventDoubleBooking, trg_CheckRoomAvailability, trg_AutoFreeRoomOnDischarge, trg_AuditBillPayment, trg_PreventMedicationDelete).

Data Import/Export & Utilities



Utility Used: SQL Server Management Studio (SSMS)
Import Flat File Wizard.

Execution Result: Successfully inserted external flat-file datasets (Insert Data: Success) into the database tables without schema errors.